# MILITARY SPECIFICATION

## PROPELLANT M1 FOR USE IN 90MM CARTRIDGES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers M1 Propellant for use in Cartridges, HE-T, M71A1 and TP-T, XM764 for M36 and M41 guns.

1.2 Classification. The propellant shall be cylindrical multiperforated grains.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-652 - Propellants Solid for Cannons Requirements and Packing

DRAWINGS

ARMY

9297850 - Cartridge, 90MM, TP-T XM764
8849017 - Cartridge, 90MM, HE-T, TP-T, M71A1 Assembly

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

FSC: 1375

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CODE OF FEDERAL REGULATIONS

Title 49, Transportation: Parts 0-199

AMC PAMPHLET 706-114

Experimental Statistics, Section 5, Tables

The Code of Federal Regulations is available from the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders for the above publication should cite: 49 CFR 0-199 (latest revision).

## 3. REQUIREMENTS

3.1 Materials. The propellant shall be M1, conforming to the requirements of MIL-STD-652.

3.2 Proving Ground Assessments. The propellant when fired in the M41 Gun and with the weight of projectile (M71, 23.4 lb) for which it was manufactured, shall comply with the following requirements when tested as specified in 4.4.1.

3.2.1 Charge Establishment. At 70 degrees Fahrenheit (°F) and standard weapon conditions, the propellant shall be capable of assessment to a velocity of 2400 feet per second (fps) without an associated expected pressure greater than 45,600 pounds per square inch (psi).

3.2.2 Uniformity of velocity. The standard deviation of velocity for the lot at 70°F shall not exceed 6.0 fps.

3.2.3 Permissible Individual Maximum Pressure. The propellant at recommended charge weight shall not produce a chamber pressure greater than 54,000 psi at any operating temperature (-50°F to +145°F) when fired in a standard weapon.

3.3 Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. No reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification unless otherwise specified.

## 4. QUALITY ASSURANCE PROVISIONS



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4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections. Quality Conformance Inspection shall be conducted on this item.

4.3 Quality Conformance Inspection

4.3.1 Inspection Lot Formation. Inspection lots shall comply with the lot formation provision of MIL-A-48078.

4.3.2 Examination and Testing

4.3.2.1 Materials (see 3.1). Each lot of propellant shall be sampled and tested for composition, form, dimensions and chemical physical properties in accordance with MIL-STD-652.

4.3.2.2 Proving Ground. (see 3.2) Each lot of propellant shall be sampled for proving ground tests in accordance with MIL-STD-652 except that ten containers shall be sampled for both initial and subsequent production lots.

4.3.2.2.1 Charge Establishment (see 3.2.1). The lot shall be rejected if the assessed charge weight of propellant has an expected average pressure which does not meet the applicable requirement.

4.3.2.2.2 Uniformity

4.3.2.2.2.1 Velocity (70°F) (see 3.2.2). If the product of the factor (see 6.3) times the velocity standard deviation of the rounds tested in accordance with 4.4.1.2 exceeds the applicable requirement the lot shall be rejected unless there is a outlier (see 6.10) in which case the lot shall be retested.

4.3.2.2.2.2 Pressure (see 3.2.3)

4.3.2.2.2.2.1 Initial Production. If any individual pressure observed during uniformity firings at any operating temperature after correction for difference between as fired and recommended service charge weight exceeds the applicable requirement the lot shall be rejected.



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Inactive for New Design. The compiled series shall be

4.3.2.2.2.2.2 Subsequent Production. If any individual pressure observed during the 70°F uniformity series when corrected to standard conditions and recommended charge weight exceeds 54,000 psi minus the quantity 75 times the pressure/temperature coefficient established during Initial Production Testing, the lot shall be tested at +145°F in accordance with 4.4.1.2.1. If any individual pressure at 145°F, when corrected to standard conditions and recommended charge weight exceeds 54,000 psi the lot shall be rejected.

4.4 Test methods and procedures

4.4.1 Proving ground assessments. These tests shall be conducted at a Government proving ground (designated by the contracting officer) in accordance with the applicable Proving Ground Acceptance Test Procedure.

4.4.1.1 Charge Establishment

4.4.1.1.1 Charge Slope Method. Three or more groups of rounds loaded with selected charge weights of propellant (weights selected should bracket expected service charge weight) shall be conditioned and fired at 70°F. From the data obtained a charge weight shall be calculated which will yield the prescribed service velocity (see 3.2.1). This calculated charge weight shall then be loaded into rounds for the uniformity series (see 4.4.1.2). The difference between the corrected velocity obtained during the uniformity series and the service velocity shall be compensated for by a final adjustment in charge weight using the slope of the velocity charge weight curve previously established. If the final adjustment in charge weight exceeds 2 percent of the weight used in uniformity firngs, the lot shall be retested for charge establishment (see 6.9).

4.4.1.1.2 Constant Slope Method. After ten (10) lots of a given production series have been fired to provide a good knowledge of the propellant characteristics, the charge slope method of assessment may be discontinued and the charge weight determined by the constant slope method. A composite charge weight versus velocity and pressure curves can be established from test results generated. This curve can then be used as in 4.4.1.1.1 above to calculate the charge weight needed to achieve service velocity. This calculated charge weight is then to be loaded into rounds for the uniformity series and final adjustment is to be made as indicated above for the charge slope method. If the final charge weight adjustment exceeds 2 percent of the weight used in uniformity firings, the lot shall be retested for charge establishment in accordance with 4.4.1.1.1 (see 6.9).



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4.4.1.2 Uniformity

4.4.1.2.1 Initial Production. Thirty rounds shall be assembled with the established charge weight of propellant. Ten rounds each shall be conditioned to -50°F, 70°F and 145°F (as indicated by thermocoupled control round). Rounds fired at 70°F shall be fired alternately with calibration rounds and shall be corrected to standard weapon/firing conditions (i.e. the difference between calibration performance under test conditions and standard conditions). Firing results at all temperatures shall be corrected to standard projectile weight (see 3.2) and to recommended charge weight conditions (i.e. the difference between as fired charge weight and recommended charge weight). The means and standard deviation of the velocities and pressures at all firing temperatures shall be calculated using corrected results. The average pressure-temperature coefficient for the lot shall be calculated from the pressure data obtained at 70°F and 145°F. The average of the coefficients for ten lots shall be utilized to calculate a 70°F upper pressure control for use in subsequent production.

4.4.1.2.2 Subsequent Production. Ten (10) rounds shall be assembled and tested at 70°F, fired alternately with calibration rounds. The means and standard deviations of the velocity and pressure shall be calculated on results corrected to standard projectile weight (see 3.2), standard weapon/firing conditions and recommended charge weight conditions. If any individual pressure exceeds the calculated 70°F upper pressure control (see 4.3.2.2.2.2.2) ten additional test rounds shall be conditioned and fired at 145°F. All results at 145°F shall be corrected to standard projectile weight (see 3.2) and recommended charge weight conditions. If any individual round at 145°F exceeds 54,000 psi, the lot shall be rejected.

4.4.1.3 Retest. If for any reason the proving ground considers that test conditions have detrimentally affected the test results, additional charges as required shall be tested. A lot shall be retested whenever any of the following conditions exist and whenever the proving ground deems it necessary.

a. Calibration rounds fired in any test phase fail to comply with Section 3 of this specification.

b. Calibration correction applied exceeds two percent of the expected muzzle velocity.



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c. If calibration components are available, but not utilized in both test and calibration rounds (see 6.9)
d. Any test round result is an outlier (see 6.10)

4.4.1.4 Referral. A test lot shall be referred to Picatinny Arsenal, ATTN: SARPA-QA-A-P for disposition whenever any of the following conditions exist and whenever the proving ground deems it necessary.
a. Number of test or calibration rounds and/or results utilized is less than specified.
b. Proper Calibration components are not available.

## 5. PREPARATION FOR DELIVERY

5.1 Packing and marking. Packing and marking shall be in accordance with MIL-STD-652.

## 6. NOTES

6.1 Ordering Data. Ordering data shall be in accordance with MIL-A-48078(PA) with the following exceptions.

6.1.1 Procurement Requirements. Procurement requirements of MIL-A-48078(PA) shall apply except for First Article Sample.

6.1.2 Contract Data Requirements. In addition to the normal distribution of records, when the propellant is procured by the Department of the Army, one copy of all ballistic test data and one copy of the Propellant Description sheet shall be forwarded to: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-P, and one copy to ATTN: SARPA-AD-E-P-2, Dover, New Jersey 07801.

6.2 Standard deviation. The standard deviation shall be calculated with (n-1) as the devisor in a standard statistical technique equivalent to that shown in MIL-STD-414, Section B, Example B-1.

6.3 Factors. The factors, as taken from Table A-21 of AMC Pamphlet 706-114 using the 95 percent upper confidence limits (A.05) and n-1 degrees of freedom, make allowances for the probability that a standard deviation from a smaller sample size exceeds the true standard deviation by chance alone.

6.4 Initial production lots. The initial production lots represent the first ten consecutive lots from each manufacturer which have been produced and met the acceptance criteria of this specification. Lapse in production of 90 days or more shall cause testing to revert back to initial production plan.



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6.5 Tolerance. A tolerance of -5°F applies to all -50°F and 145°F requirements and a tolerance of plus or minus 2°F to all 70°F requirements.

6.6 Surveillance sample. Samples shall be forwarded to Picatinny Arsenal, Dover, New Jersey, ATTN: SARPA-FR-G for the propellant surveillance test program in accordance with MIL-STD-652.

6.7 Test Charges. Test charges shall be assembled in accordance with drawing 8849017.

6.8 Proving ground test summary

Initial Production

| Temp (°F) | Sample Size | Vel(fps) (assessed) | Lot Std. Dev of Vel (fps) | Perm Ind. Max Pres (psi) | Avg. Pres. Max (psi) |
|---|---|---|---|---|---|
| -50 | 10 | NA | NA | 54,000 | NA |
| 70 | 10 | 2400 | 6.0 | 54,000 | 45,600 |
| 145 | 10 | NA | NA | 54,000 | NA |

Subsequent Production

| | | | | | |
|---|---|---|---|---|---|
| 70 | 10 | 2400 | 6.0 | see 4.3.2.2.2.2.2 | 45,600 |
| 145* | 10 | NA | NA | 54,000 | NA |

*To be fired only when 70°F results indicate that marginal pressure performance may exist (see 4.3.2.2.2.2.2).

6.9 Retest. All original and retest data shall be submitted to Picatinny Arsenal, ATTN: SARPA-QA-A-P, Dover, New Jersey 07801.

6.10 Outlier. The following test shall be performed to determine if a single test result is an outlier. With a sample



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size of 10 at a significance level of 0.05, the critical value ($T_c$) is 0.477. Calculate a test value ($T_t$) as follows:

For Low Outlier

$$T_T = \frac{(X_2 - X_1)}{(X_9 - X_1)}$$

For High Outlier

$$T_T = \frac{(X_{10} - X_9)}{(X_{10} - X_2)}$$

Where:

$X_1$ = test result being tested as low outlier
$X_2$ = test results adjacent to $X_1$, after all the test results are ordered from smallest to largest
$X_9$ = test results adjacent to the largest test result
$X_{10}$ = test result being tested as high outlier

$X_1$ or $X_{10}$ is an outlier if $T_t$ is greater than $T_c$.

Figure 1 presents a schematic of the decision process imposed when the outlier test is utilized.

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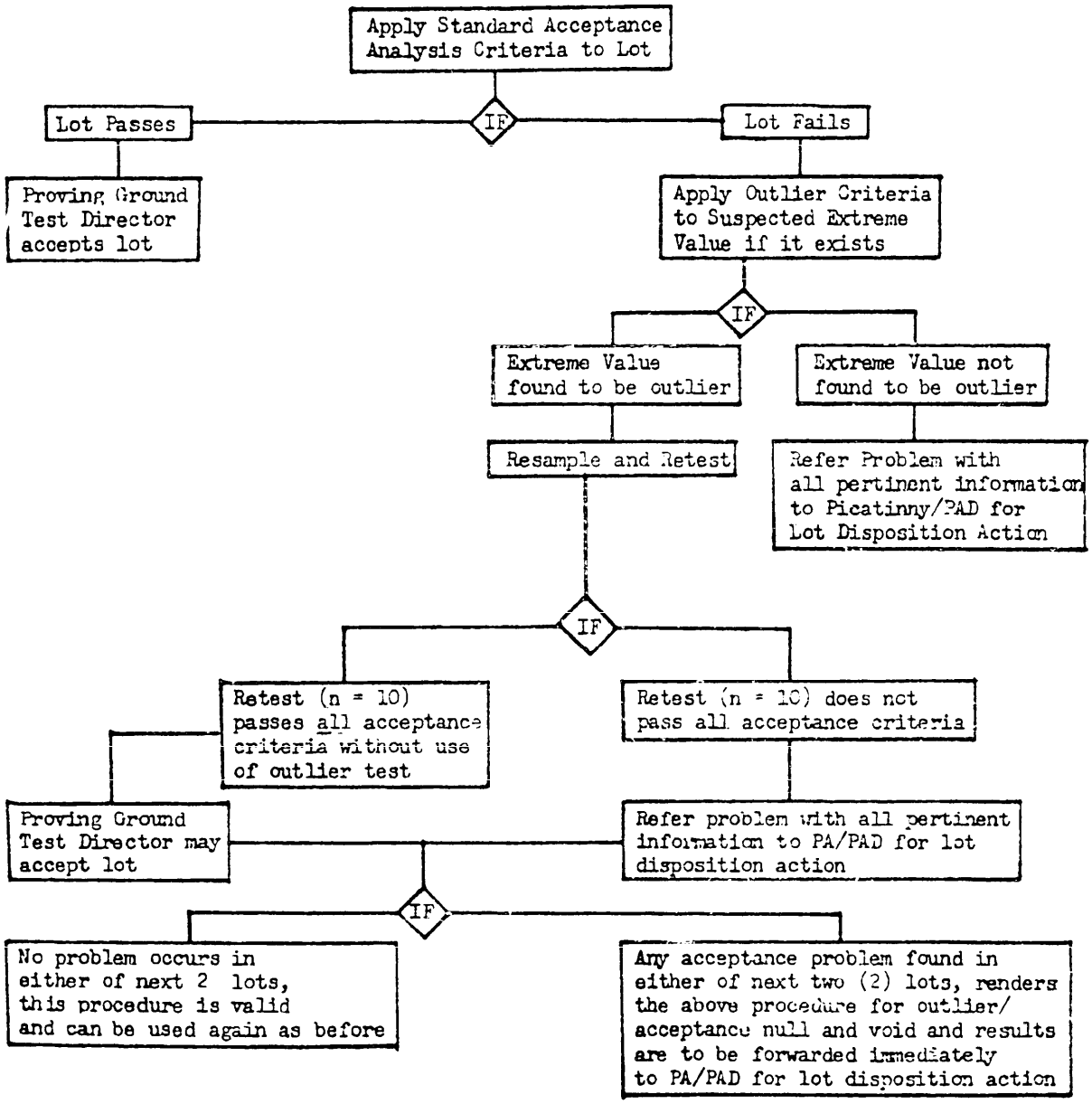
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FIGURE 1. DECISION PROCESS WITH OUTLIER TEST






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